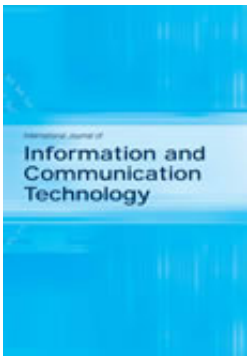
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A physics-informed neural network-based edge-cloud collaborative digital twin for predictive maintenance of ageing railway signalling equipment

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Abstract: This study proposes a physics-informed neural network (PINN)-based edge-cloud collaborative digital twin (DT) for predictive maintenance of ageing railway signalling equipment (RSE). The core model, an adaptive pigeon-inspired optimised PINN (API-PINN), integrates electromechanical degradation physics with operational data to forecast faults and estimate remaining useful life (RUL). The architecture enables low-latency, real-time anomaly detection at the network edge while leveraging the cloud for intensive model training. Experiments on real-world railway signalling data sets can show that API-PINN attains the prediction accuracy of 98.77% and the R^2 of 0.968, which is better than the baseline models. The given framework offers a strong performance when the data is noisy and imbalanced, allowing for the arrangement of the maintenance schedule proactively, minimising the costs of operations, minimising downtimes, and improving safety.

Keywords: predictive maintenance; digital twin; edge-cloud computing; railway signalling equipment; RSE; ageing infrastructure.

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1 Introduction

Ageing railway signalling equipment (RSE) is defined as inheritance electromechanical or electronic control systems that have lost performance, module degradation, and reliability as a consequence of long-term stress of use and experience to environmental fundamentals. These degradations result in disorganisations, operative delays, and amplified incident risks. As quantified in Ji et al. (2025) and Wang et al. (2024b), the transformation of RSE has developed a necessity to preserve security and efficiency.

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Examples of difficulties are outmoded technologies that are not able to support modern operational needs, connotations that are not well modified to real-time traffic circumstances, and the wear of the machineries, including microelectronic wear. Railway systems are using large-scale asset substitution, preventative repairs, and data-based dependability techniques to solve these problems. The recent introduction of solid-state interlocking (SSI), light-emitting diode (LED) signals that are and constantly track circuiting has made things more durable and less likely to break. Data loggers also let you keep an eye on how well your equipment is working all the time so you can do predictive maintenance. Cybersecurity features are also being added to networks more and more to keep them safe from digital attacks (Wang et al., 2024c; García-Méndez et al., 2025; Vieira et al., 2024).

The perpetual modification in RSE structure is characterised by technological innovation and robotics. Contemporary signalling technology based on electronic interlocking (EI) and automatic train control (ATC) is lowering human participation and cumulative operating efficiency (Drózd and Rosiński, 2023). Communication-based train control (CBTC) is a well-known concept in metropolitan systems that lets block trains through by using wireless interaction all the time, which increases track capacity and operational reliability. In addition, train control systems (TCS) and positive train control (PTC) use global positioning system (GPS) and on-board detectors for interconnection and collision avoidance (Atik et al., 2025; Sun et al., 2019; Liu et al., 2024). The junction of internet of things (IoT) devices, artificial intelligence (AI), and big data statistical analyses empowers real-time surveillance, fault forecast, and adaptive conservation development. The arrival of 5G and fibre-optic communication systems could only make the data broadcast speed faster, permitting quicker making decisions in real-time traffic administration systems. Even with these expansions, there are still problems to solve, such as the excessive cost of converting infrastructure, the difficulty of integrating old systems, worries about interoperability, the risk of cyberattacks, and the lack of trained staff who know how to use modern digital signalling systems (Fernandes et al., 2025; Ibadah et al., 2024; Harrod, 2025; Chacón et al., 2024). This change is a compromise between moving to smarter, data-driven operational models and keeping costs and integration in mind. Comparative analyses ranging from electromechanical signaling to digital control architectures incorporating IoT sensors and predictive dashboards indicate a transition toward intelligent, autonomous railway systems.

To address these issues, a physics-informed neural network (PINN)-based edge-cloud collaborative digital twin (DT) architecture (Kazemi Naeini et al., 2025) has been developed. Thus, it is a hybrid approach that combines physics-based modelling with machine learning (ML) that allows real-time emulation, monitoring, and predictive analysis of signalling equipment. The framework can be broken down into three layers: physical asset layer (IoT sensors for data collection), edge layer (localised devices like FPGAs and microcontrollers running lightweight PINN models for low-latency anomaly detection), and a cloud layer (computationally intensive PINN models for model refinement and system-level optimisation) (Yang et al., 2024; Zia et al., 2025). The PINN structure encodes physical laws at the neural level and requires less data, and makes appropriate predictions even if there is not much data. The edge-cloud sync mechanism ensures a smooth data flow, where the edge performs instantaneous calculations and the cloud updates every so often to enhance the precision of predictive computations. Applications of this architecture are predictive maintenance of aircraft systems, electric vehicle (EV) batteries, smart grid energy management, energy-efficient robotic control,

etc. The edge intelligence and cloud learning are in balance, and both together ensure their advantages of low latency, high accuracy, and scalability, which lay a sophisticated and efficient basis for predictive maintenance and operation optimisation in modern railway signalling infrastructures (Barimah et al., 2025; Wang et al., 2024a). This research aims to develop an intelligent DT framework that uses adaptive pigeon-inspired optimised physics-informed neural network (API-PINN) and edge-cloud collaboration to predict and prevent failures in ageing RSE. It aims to enable real-time monitoring, early fault detection, and proactive maintenance, thereby improving safety, reliability, and cost-efficiency in railway operations.

2 Literature review

Current research in autonomous railway preservation employs ML, deep learning (DL), along with decision trees (DT) to improve the ability to find problems, make predictions, and make decisions based on data. This integration has led to more accurate diagnoses, less frequent maintenance, and a more reliable system. Reinforcement learning (RL) and unsupervised detection of anomalies are two important techniques that have problems with scalability and real-time flexibility. This shows that a physics-based strategy is needed to manage ageing railway signalling systems. The time-series current data-driven fault diagnosis (TSCD-FD) technology (Sugiana et al., 2023) improves the maintenance of railway point machines (RPMs) by removing features from the time domain. It does this by using DT and random forests (RF), which are lucrative and more accurate, but they can only be used on convinced types of RPMs. The vibration-based signal-based fault detection and identification (VS-FDI) procedure (Sun et al., 2023) merges energy randomness with time-frequency material, employing support vector machine (SVM) classifiers, its efficacy declines in noisy environments. The enhanced deep convolutional neural network-support vector data description (DCNN-SVDD) outline (Shi et al., 2023), employing advanced DL algorithms, achieves 96.59% accuracy in fault recognition for ZDJ7 machineries but incurs significant training difficulty. The wavelet denoising-based vibration monitoring (WD-VM) method also finds squat difficulties in railway switches and adventures early on. It finds links between quivering and squat importance, but sensitivity drops off at a convinced distance (Zuo et al., 2023). The acoustic feature-based ensemble fault detection (AFE-FD) method (Rustam et al., 2023) uses Chi-square feature assortment and mel frequency cepstral coefficients (MFCCs) to find failures in railway tracks, it works better with 60 features. Even so, its performance gets worse because of noise in the environment and changes in the dataset. The actor-critic through the advantages digital twin-based predictive maintenance (A2C-DT-PM) organisation (Sresakoolchai and Kaewunruen, 2023) uses deep RL and DT technology to look at four years of railway data and cut conservation work by 21% and problems by 68%. But it is hard to scale and needs a lot of computing influence. The data-driven unsupervised predictive maintenance (DD-UPM) context (Meira et al., 2023) for air industrial system uses half-space trees and one-class k-nearest neighbour to invention irregularities in real time, although it has trouble adjusting to different data distributions. The double-scale wide first-layer kernel convolutional neural network (DS-WCNN) model (Chen et al., 2023) offers inexpensive vibration-based fault judgement for RPMs by employing low-sampling-rate sensors and dual timescale feature extraction, therefore improving

accurateness and noise resilience. But there are still difficulties with being able to adjust in real time and move to another RPM models.

The climate-aware ML-based clustering (CMLC) procedure (Kasraei and Garmabaki, 2024) groups switches and crossings (S&C) in the railway network based on historical and climate data. It finds four climate-based groups to help with maintenance planning, but it does not work well in different places or with long-term climate changes. The A data-driven multilabel maintenance detection (DD-MMD) method (Ghiasi and Malekjafarian, 2025) identified railway track required maintenance from train momentum data, obtaining 95% accuracy through time- frequency data extraction and analysis of variable (ANOVA)-mediated feature selection, but effectiveness shift with varying sensor setups alongside track conditions. Finally, the ML-based tie condition estimation (ML-TCE) framework (Khademi et al., 2025) uses Canadian data to predict faulty railway ties. It works best with RF models for scheduling maintenance, although different rail systems need to be recalibrated.

To provide a clear and concise overview of how our proposed API-PINN framework advances the state-of-the-art, we have summarised the key differences between our approach and the related works discussed in Table 1.

Table 1 Comparative analysis of existing railway fault detection approaches vs. proposed API-PINN Framework

Author/model	*Methodology*	*Strengths*	*Limitations*
Sugiana et al. (2023) (TSCD-FD)	Decision trees, random forest	High accuracy for RPMs	Limited to specific RPM types
Sun et al. (2023) (VS-FDI)	VMD, ReliefF, SVM	Effective feature selection	Poor performance in noisy environments
Shi et al. (2023) (DCNN-SVDD)	Deep CNN, SVDD	96.59% accuracy	High computational complexity
Zuo et al. (2023) (WD-VM)	Wavelet denoising	Early fault detection	Reduced sensitivity at distance
Rustam et al. (2023) (AFE-FD)	MFCC, ensemble learning	60-feature optimisation	Environmental noise sensitivity
Sresakoolchai and Kaewunruen (2023) (A2C-DT-PM)	Deep RL, digital twin	68% fault reduction	Scalability issues, computationally intensive
Meira et al. (2023) (DD-UPM)	Half-space trees, k-NN	Real-time anomaly detection	Poor adaptation to data distribution changes
Chen et al. (2023) (DS-WCNN)	Dual-scale CNN	Noise resilience, low sampling rate	Limited real-time adaptability
Zhou et al. (2021) (GC-BiLSTM-CRF)	Graph CNN, BiLSTM	High entity recognition	Text-based only, no physics
Yang et al. (2025) (BERT-BiLSTM-CRF)	Transformer, BiLSTM	98.72% entity recognition	Post-hoc analysis only
Proposed API-PINN	PINN + APIO + Edge-Cloud DT	98.77% accuracy, $R^2 = 0.968$, 47.2% downtime reduction	Requires high-quality operational data

While existing research has explored various ML approaches for railway fault detection – such as TSCD-FD (Sugiana et al., 2023) for point machines, DCNN-SVDD

(Shi et al., 2023) for ZDJ7 equipment, and A2C-DT-PM (Sresakoolchai and Kaewunruen, 2023) for maintenance optimisation – these approaches are predominantly data-driven and lack integration with physical degradation principles. The proposed API-PINN framework introduces three key novelties:

1 Physics-informed degradation modelling that embeds electromechanical laws (Kirchhoff's Voltage Law, thermal drift characteristics, and relay circuit dynamics) directly into the neural network architecture, ensuring physically consistent predictions even with limited fault data.

2 Adaptive pigeon-inspired optimisation (APIO) with dynamic parameter tuning that outperforms standard metaheuristics in convergence speed and solution quality for nonlinear, time-varying railway signalling systems.

3 Edge-cloud collaborative architecture specifically designed for ageing infrastructure, enabling sub-second anomaly detection at the edge while continuously updating physics-based models in the cloud – a capability absent in existing centralised or purely edge-based approaches.

3 Methodology

Ageing RSE is a highly coupled electromechanical-cyber system consisting of programmable logic controllers (PLCs), relays, sensors, and control logic circuits for coordinating the movement of trains in a fixed block signalling (FBS) network. Each track block is isolated and controlled by a PLC-based block controller that enforces train separation through sequential signal actuation governed by inductive sensors, axle counters, and timed relay logic. In this setup, electrical excitation of the relays with logical feedback loops from track circuits determines the signal occupancy states, such as red, yellow, and green, to ensure safe train spacing under operational constraints. These PLC-based block controllers represent the physical layer of the DT, in which continuous mirroring of electrical, thermal, and logical states occurs virtually for real-time condition assessment of the system. The architecture of the ATC and automatic train protection (ATP) subsystem extends to include sensor pairs (n, n + 1), delay timers such as timer ON-delay (TON), timer coil holding (TCH), and Jack Kilby (JK) flip-flop circuits to encode the logical transitions of block occupation and clearance. During long operational lifecycles, contact wear, oxidation-induced resistance drift, and coil degradation in the signalling components lead to several symptoms, including signal delays, spurious triggering, and incomplete relay actuation, which are important indicators for predictive maintenance.

The degradation behaviour of the electromechanical components can be captured by a physics-informed degradation state model, rate of change of the instantaneous health state $(\frac{dX(t)}{dt})$ expressed in equation (1).

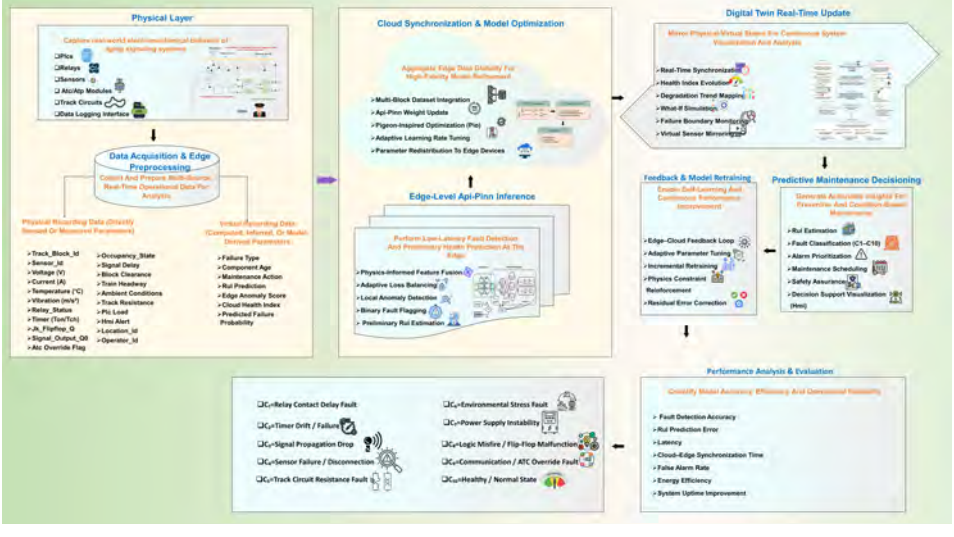
$$\frac{dX(t)}{dt} = -\alpha X(t) + \beta I(t) + \eta(t) \tag{1}$$

where $X(t)$ represents the instantaneous health state of a signalling component (e.g., relay coil efficiency or signal delay response), α is the degradation coefficient representing wear or contact resistance rate, β is the electromechanical gain factor linking PLC excitation current $I(t)$ to response performance, and $\eta(t)$ represents a stochastic perturbation term accounting for environmental effects such as vibration, humidity, or thermal stress. The observable system degradation output $Y(t)$ is composed of both physics-based and data-driven components, expressed as equation (2).

$$Y(t) = f_{phys}\left(X(t), \theta_{phys}\right) + f_{DL}\left(X(t), \theta_{DL}\right) + \in \tag{2}$$

where $f_{phys}(\cdot)$ is the physics-informed analytical model derived from relay circuit equations and thermal drift characteristics, $f_{DL}(\cdot)$ symbolises the data-driven modification term refined through the API-PINN, θ_{phys} and θ_{DL} are model strictures, and $\in$ is the residual noise term reflecting number uncertainty. The API-PINN takes gain of this hybrid building to fuse physical degradation laws with historical operative data, thereby producing interpretable and high-fidelity maintenance predictions. The model is uninterruptedly modified by using a pigeon-inspired swarm optimisation procedure that updates the neural strictures to minimalise both the data loss and physics equilibriums in time-varying functioning conditions. In this research, the proposed model workflow functionality is depicted in Figure 1.

Figure 1 Proposed model functional flow diagram (see online version for colours)



The edge-cloud collaborative architecture comprises an edge layer that consists of distributed PLCs and local sensor nodes, which are responsible for collecting real-time voltage, current, and occupancy data. In turn, these edge units will conduct low-latency anomaly detection and preliminary API-PINN inference that will trigger maintenance warnings locally. It also contains a Cloud Layer that acts as the global synchronisation and learning core by aggregating multi-block datasets, retraining the API-PINN with updated physical parameters, and redistributing optimised weights to the edge devices. The supervisory level consists of a human-machine interface (HMI) that displays

synchronised DT data on block occupation, signal transition states, ATC override states, and predictive alarms on component degradation. Thus, this integrated DT framework forms a self-evolving, physics-informed cyber-physical system that is capable of continuous monitoring, adaptive fault prediction, and data-driven maintenance scheduling. Therefore, it modernises the legacy RSE infrastructures while ensuring operational reliability and safety.

3.1 Real-time PLC-based railway signalling health and fault prediction dataset

The dataset is collected from a Kaggle (https://www.kaggle.com/datasets/programmer3/real-time-railway-signalling-fault-dataset) railway atmosphere to mimic the function of an industrial-grade monitoring system for RPMs and FBS networks. Every record has timestamped data from PLC parts, track monitors, and computational units. This data shows mechanical, electrical, environmental, and operational shifts that are important for finding faults, analysing anomalies, and estimating remaining useful life (RUL). Some of the most important characteristics are voltage, current, temperature, vibration, relay, clock operations (TON, TCH), JK flip-flop outputs, ATC flags, and occupant states. The degradation modelling is helped by other parameters including signal transmission delay, train headway, and considerations about the environment. Operational diagnostics use the CPU load of the PLC, warning codes from the HMI, and types of failures. Element age, service activities, and RUL projections all help with life-cycle management. Before cloud synchronisation, procedures like filtering and standardisation are done to the data. The API-PINN combines criteria for physical deterioration with real-time analytics to improve failure prediction, restoration scheduling, and infrastructure reliability. Identifiers make it possible to trace faults in ageing signalling systems correctly by providing context.

3.2 API-PINN model for intelligent fault prognostics and RUL estimation

The API-PINN model combines RSE physics and swarm-based optimisation to make sure that the predictions are accurate and physically consistent. It incorporates electromechanical partial differential equations of relay circuits, vibration, and thermal drift into the neural structure, and the PIO layer simulates the navigation of pigeons to balance the weights of the neural network at a global level, avoiding local minima and slow convergence. The adaptive control dynamically adjusts the learning parameters from the system feedback, and the learning parameters can converge quickly under different conditions. Together, these mechanisms allow for the ability to model degradation accurately, predict faults, and estimate the RUL in ageing signalling systems.

3.2.1 PINN formulation for fault diagnosis and predictive maintenance

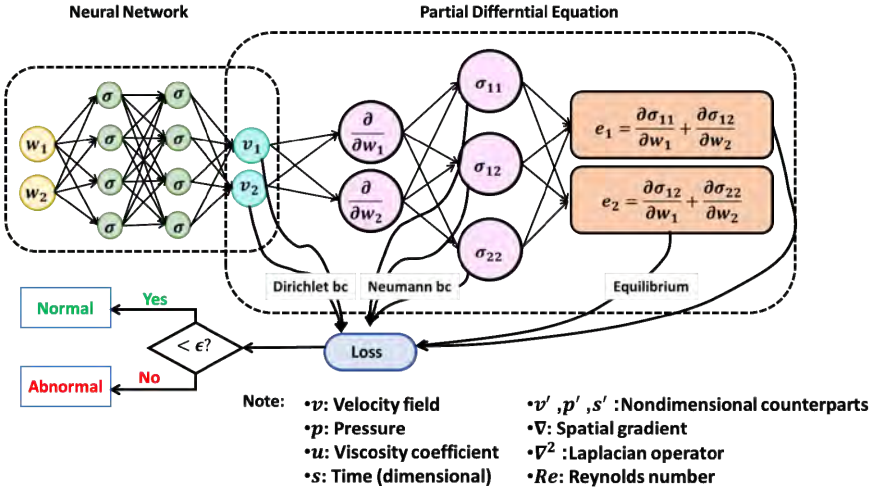
A PINN is a kind of DL model that incorporates differential equations to include physical rules into its loss function. In this paper, the PINN is designed to embed the physical rules governing how PLC-based RSE interact with both electricity and signals directly into the learning process. The PINN is not only fitting data but also including physics restrictions on the loss function such that fault prediction and degradation modelling are physically consistent. Figures 2(a) and (b) shows how the API-PINN-based edge-cloud DT learns how to provide predictive maintenance on old RSE. It begins with partial

differential equation (PDE) non-dimensionalisation for stable electromechanical modelling, and then it moves on to a Fourier feature network that uses random weight factorisation and boundary conditioning. It makes sure that fault classification is based on equilibrium by using loss balancing, causal training, and curriculum training. This lets it tell the difference between normal and abnormal states for accurate real-time fault detection and RUL prediction.

Figure 2 Schematic representation of, (a) PDE training workflow is causal learning optimisation (b) PINN architecture for predictive maintenance tool (see online version for colours)



(a)



(b)

The first step in the workflow is to non-dimensionalise the PDE and construct the network architecture (including Fourier feature networks and boundary conditions). The second step is to use a training method that balances loss, uses causal learning, and follows a curriculum. A nonlinear PDE system can show how voltage, current, and temperature change over time in relays and track circuits, system performance over all time $(\frac{\partial u}{\partial t})$ as in equation (3).

$$\left(\frac{\partial u}{\partial t}\right) + N[u] = 0, t \in [0, T], x \in \Omega \tag{3}$$

where $u(t, x)$ signifies the system state factors such as coil current, relay voltage, or temperature distribution) in different time domain (t) and spatial domain (x), and $N[\cdot]$ describes a nonlinear differential operator that understands the interaction between electrical and mechanical systems, signal delay, time horizon (T) and degradation effects in the system domain Ω. The system is governed by boundary and beginning conditions delineated as equation (4).

$$u(0, x) = g(x),\ B[u] = 0 \tag{4}$$

where $g(x)$ signifies the preliminary sensor state distribution, and $B[\cdot]$ is the boundary operator, which can be either Dirichlet (fixed potential/temperature) or Neumann's (flux-based), depending on the circuit's limits. The neural network learns an estimate u_θ of $\theta(t, x) \approx u(t, x)$, where θ symbolises the trainable network restrictions. The conforming residual procedure of the PDE leading the model's physics constancy $R_\theta(t, x)$ is expressed in equation (5).

$$R_\theta(t, x) = \frac{\partial u_\theta}{\partial t} + N[u_\theta] \tag{5}$$

where $\frac{\partial u_\theta}{\partial t}$ is a time derivative of the predicted variable, u_θ represents neural network output and $N[u_\theta]$ denoted as nonlinear differential operator. The total PINN loss function ($L(\theta)$) integrates manifold physics-based constraints and is expressed as equation (6).

$$L(\theta) = L_{ic} + L_{bc} + L_r \tag{6}$$

where L_{ic}, L_{bc}, and L_r represent initial condition, boundary condition, and PDE residual losses, respectively, defined as equations (7) to (9):

$$L_{ic} = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} \left| u_\theta(0, x_i) - g(x_i) \right|^2 \tag{7}$$

$$L_{bc} = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left| B\left[u_\theta(t_i, x_i) \right] \right|^2 \tag{8}$$

$$L_r = \frac{1}{N_r} \sum_{i=1}^{N_r} \left| R_\theta(t_i, x_i) \right|^2 \tag{9}$$

where N_{ic}, N_{bc}, and N_r are the number of data points in the initial condition $g(x_i)$, boundary condition $B[\cdot]$, and PDE residual losses R_θ at spatial point x_i and time domain t_i. To expedite the convergence and identify high-frequency fluctuations in electrical fluctuations, Fourier characteristic $f(x)$ embeddings are implemented as equation (10).

$$f(x) = \left[\sin(2\pi B x), \cos(2\pi B x) \right] \tag{10}$$

where $\sin(2\pi Bx)$ and $\cos(2\pi Bx)$ are used to help the neural network capture periodic or oscillatory patterns in the railway signal data. Additionally, adaptive loss weighting L_{total} energetically regulates the comparative inspiration of physics and data terms during training, as defined in equation (11).

$$L_{total} = \sum_i \lambda_i(t) L_i, \lambda_i(t) \tag{11}$$

where L_i is an individual loss component and λ_i is denoted as an adaptive weighting coefficient, A typical illustration of a physics limitation applicable to the railway relay arrangement $(\frac{di}{dt})$ is described as equation (12).

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\, dt = V_{PLC}(t) \tag{12}$$

where L, R, and C denote the inductance, resistance, and capacitance of the relay coil or track circuit, $i(t)$ is the instantaneous current, and $V_{PLC}(t)$ characterises the PLC control current applied to the relay driver. This restriction makes sure that Kirchhoff's voltage law (KVL) is followed during the PINN learning process. This means that the predicted signal transitions must match how real-world electromechanical interactions work.

3.2.2 Pigeon-inspired optimisation for neural parameter convergence in PINN-based railway fault modelling

PIO is a swarm intelligence approach that uses the way pigeons find their way home to efficiently search for a global optimum. Its usages range and map and landmark triangulation principles. This approach progresses the trainable neural parameters (θ) of the PINN to create the PIO-PINN model, which helps the loss landscape converge quickly while still subsequent physics-consistent education subtleties. For an individual pigeon i in a D-dimensional search space, the velocity $V_i(t)$ and position $X_i(t)$ updates are administered by equations (13) and (14).

$$V_i(t) = V_i(t-1)e^{-Rt} + rand\left(X_g(t) - X_i(t-1)\right) \tag{13}$$

$$X_i(t) = X_i(t-1) + V_i(t) \tag{14}$$

where $V_i(t)$ signified as velocity of pigeon i at iteration t, $V_i(t-1)$ is a previous velocity of pigeon i represents the map–compass factor governing the deterioration of velocity with time, $X_g(t)$ is the current global best solution, and A randomised module for experiment behaviour is announced by rand(·), $X_i(t-1)$ is a previous position of pigeon i, and $X_i(t)$ denoted as Updated position of pigeon i after movement. The location update simulates progressive movement toward ideal convergence, while the velocity information simulates the pigeon's capacity to sense global electromagnetic radiation for orientation (global exploration).

To mimic landmark-based navigation, the number of pigeons decreases as the flight goes on, which speeds up convergence. Equation (15) shows how to use the population halving rule.

$$N_p(t) = \frac{N_p(t-1)}{2} \tag{15}$$

where $N_p(t)$ is the Number of pigeons in the population at iteration t, and $N_p(t-1)$ is denoted as the Number of pigeons in the previous iteration. Equation (16) uses a weighted fitness aggregation to find the centroidal landmark point.

$$X_c = \frac{\sum_{i=1}^{N_p} X_i(t) \cdot fitness\left(X_i(t)\right)}{\sum_{i=1}^{N_p} fitness\left(X_i(t)\right)} \tag{16}$$

where X_c is a central landmark or the average best position of the swarm, and $X_i(t)$ denotes the position of pigeon i at iteration t. Then, as shown in equation (17), the position of each pigeon is updated in relation to this centroid.

$$X_i(t) = X_i(t-1) + rand\left(X_c(t) - X_i(t-1)\right) \tag{17}$$

where $X_c(t)$ is the position of pigeon i in the previous iteration. The fitness function checks how well each solution works by looking at the inverse of the PINN's objective loss, as indicated in equation (18).

$$fitness\left(X_i(t)\right) = \frac{1}{f_{\min}\left(X_i(t)\right) + \in} \tag{18}$$

where $fitness(X_i(t))$ is a fitness value of pigeon i at iteration t, and $f_{\min}(X_i(t))$ is denoted as the objective or loss function value for pigeon i, and $f(X)$ represents the entire PINN loss function to be reduced as equation (19).

$$f(X) = L_{ic} + L_{bc} + L_r \tag{19}$$

The optimisation procedure ends when the difference in improvement between two iterations is less than a certain value, δ [equation (20)].

$$\left| f\left(X_{best}\right) - f\left(X_{prev}\right) \right| < \delta \tag{20}$$

where $f(X_{best})$ is the current best fitness (optimal solution) and $f(X_{prev})$ is denoted as the previous best fitness value. This termination condition ensures convergence to the ideal neuronal weights θ^*, facilitating robust and stable learning of models.

3.2.3 APIO for dynamic fault learning and real-time parameter tuning

The APIO builds on standard pigeon-inspired optimisation (PIO) by adding adaptive control parameters, variable learning speeds, and stability-related tweaking. This makes it easier for nonlinear and dynamic optimisation problems to converge faster and be more stable. APIO also deals with development delays and nonlinear fluctuations in RSE by using time-varying adaptive coefficients and new ways to start a population that change the search process in real time. The initial population version is articulated in equation (21).

$$X_i(0) = \begin{cases} K_l + (i-1)K_{step1}, & 1 \le i \le \dfrac{N_p}{2} \\ K_{initial} + IK_{step2}, & \dfrac{N_p}{2} + 1 \le i \le N_p \end{cases} \tag{21}$$

where $X_i(0)$ is initial position of pigeon i in the search space, i represent as Index of pigeon (1 to N_p), N_p signified as total number of pigeons in the population, K_l and K_u are the lower and upper boundary limit of the search region, $K_{initial}$ denoted as an

Intermediate initialisation constant defining the midpoint of search space, K_{step1} and K_{step1} are the incremental step size for pigeons in the lower half and upper half of the population, with step intermissions demarcated as equations (22) to (23).

$$K_{step1} = \frac{K_{initial} - K_I}{\left(\frac{N_p}{2}\right) - 1} \tag{22}$$

$$K_{step2} = \frac{K_u - K_{initial}}{\frac{N_p}{2}} \tag{23}$$

The velocity of each pigeon is initialised as $V_i(0) = 0$. An adaptive map-compass updating rule is presented to control velocity and acceleration based on time-varying parameter values [equation (24)].

$$V_i(t) = \alpha(t)V_i(t-1) + \beta(t)rand\left(X_g(t) - X_i(t-1)\right) \tag{24}$$

where

$$\alpha(t) = e^{-Rt\left(1-\frac{t}{T_{\max}}\right)},\ \beta(t) = \gamma\left(1 - e^{-\eta t}\right),$$

where ($\alpha(t)$) represents adaptive control functions that gradually reduce exploration, while enhancing exploitation ($\beta(t)$) over time, distance vector between the global best position $X_g(t)$ and the current particle position $X_i(t - 1)$. The adaptive landmark update guarantees that the pigeons orient themselves toward a centroid that is constantly changing, as shown in equation (25).

$$X_i(t) = X_i(t-1) + \rho(t)rand\left(X_c(t) - X_i(t-1)\right) \tag{25}$$

The constantly changing gain update law for regulation of feedback is described as equation (26) to allow for real-time control signal $u_g(n)$.

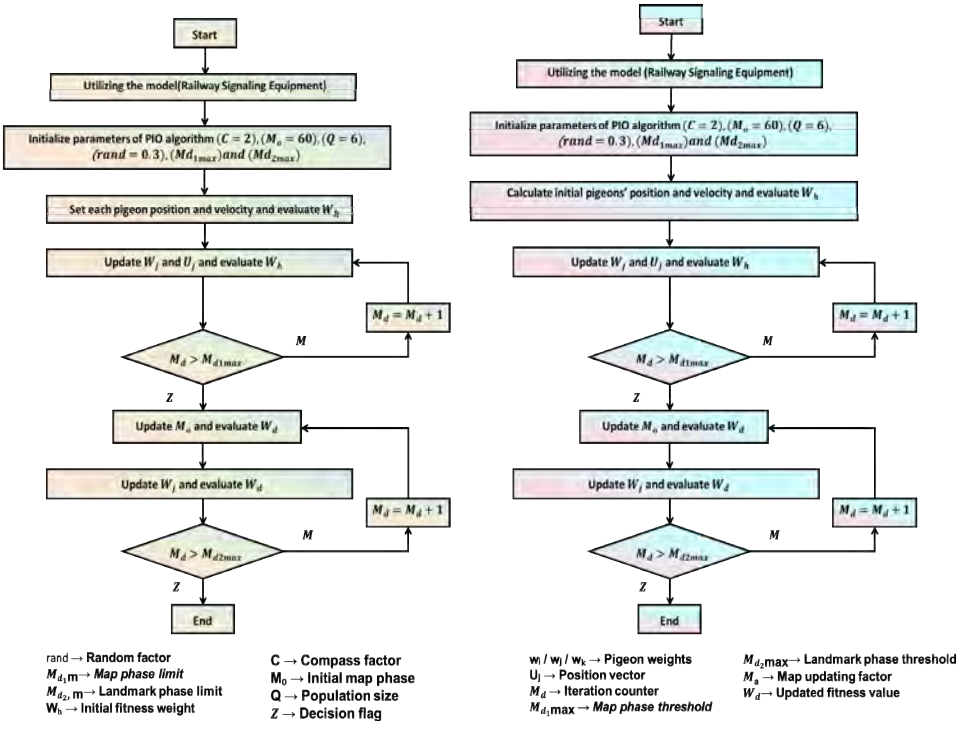
$$u_g(n) = K_q q(n) + K_I \sum_{k=0}^{n} e_{an}(k)T \tag{26}$$

where K_q and K_I are proportional and integral gains, $e_{an}(k)$ is the error signal, and T represents the sampling time, $q(n)$ is a system state deviation or feedback error at iteration n, and T is a Sampling period or time interval. Equation (27) shows how to balance energy use and error reduction in a multi-objective cost function.

$$J = \int_0^{\infty} \left(k_1 q^2 + k_2 u_g^2 + k_3 e_{an}^2\right) dt \tag{27}$$

where J is a performance index or cost function representing total optimisation error, k_1, k_2, and k_3 are the Weighting coefficients for charge (q), control (u_g), and error energy (e_{an}) terms in rate of change of time period (dt) contribution from one variable (like tracking error or system state).

Figure 3 Flowchart, (a) predictive maintenance decision using PIO (b) anomaly and RUL classification using API (see online version for colours)



For transient stability control, rise-time adjustment and overshoot penalty factors are introduced, as shown in equation (28).

$$J = J + k_4 t_r + k_5 \left| e_{an}(t) \right| \tag{28}$$

where t_r is a rise time, representing system transient response duration, k_4 and k_5 are the overshoot and steady-state penalty coefficients, and $|e_{an}(t)|$ is an absolute instantaneous adaptive error magnitude. The overall computational complexity (O) of APIO is articulated as equation (29).

$$O\left(DN_p N_{c1}^{\max} + \left(DN_p + N_p^2\right) N_{c2}^{\max}\right) \tag{29}$$

where D is the dimensionality of the search space, N_p is signified as the total pigeon population, $N_{c1}^{\max}$ and $N_{c2}^{\max}$ are the maximum iteration counts in phase 1 and phase 2 of APIO. Finally, the adaptive boundary adjustment maintains stability in the dynamic search space as in equation (30).

$$K_p : 0 \to 1, K_\phi : 0 \to \left(25 - L_p\right) K_p + L_p L_n \tag{30}$$

where K_p is an adaptive proportional scaling factor for dynamic boundary adjustment, K_ϕ is an adaptive boundary modulation parameter controlling exploration-exploitation balance, L_p denoted as landmark learning phase parameter, L_n signified as normalisation constant for adaptive limit scaling. In the context of the RSE with DT system, APIO's

adaptive updates provide exact neural parameter convergence and stable performance when learning by dynamically balancing exploration (global search) and exploitation (local refining). The PIO and API optimisation loop in Figure 3(a) and Figure 3(b) shows how weights are updated and fitness is evaluated over and over again. Lastly, the system sorts signalling circumstances into normal and abnormal categories. This lets it find anomalies, sort faults, and predict RUL in real time, which helps with proactive maintenance decision-making.

3.2.4 Parameter selection and sensitivity analysis

To ensure the reproducibility and robustness of the API-PINN model, this section details the initialisation and selection of its key parameters. The APIO algorithm's performance is governed by several critical parameters. The map and compass factor R in the velocity update equation was initialised at 0.3, based on preliminary experiments that balanced global exploration in early iterations with local exploitation in later stages. The maximum number of iterations for the map and compass phase ($N_{c1}^{\max}$) and the landmark phase ($N_{c2}^{\max}$) were set to 100 and 50, respectively, ensuring sufficient convergence. The population size (N_p) was set to 50, a value that provided a good trade-off between computational efficiency and search diversity. For the adaptive control, the proportional gain (K_q) and integral gain (K_I) in the control signal $u_g(n)$ were tuned using a heuristic approach to minimise the cost function J, with final values of 0.5 and 0.1, respectively.

A sensitivity analysis was conducted on the two most influential adaptive parameters: $\alpha(t)$ and $\beta(t)$ from equation (24). $\alpha(t)$ controls the inertial influence of a pigeon's previous velocity, while $\beta(t)$ scales the attraction to the global best position. Varied the base constants γ and η within $\beta(t)$ over a range of ±50% from their nominal values (γ = 1.5, η = 0.01) and observed the impact on the model's convergence speed and final loss. The results, summarised in Table 2, indicate that the API-PINN maintains stable performance across a wide range of these parameters, with the R^2 score for RUL prediction varying by less than 0.5%. This robustness is attributed to the adaptive nature of the update rules, which dynamically correct for sub-optimal initial parameter choices based on real-time system feedback. This stability is critical for deploying the model in diverse and changing operational environments without the need for frequent, laborious manual re-tuning.

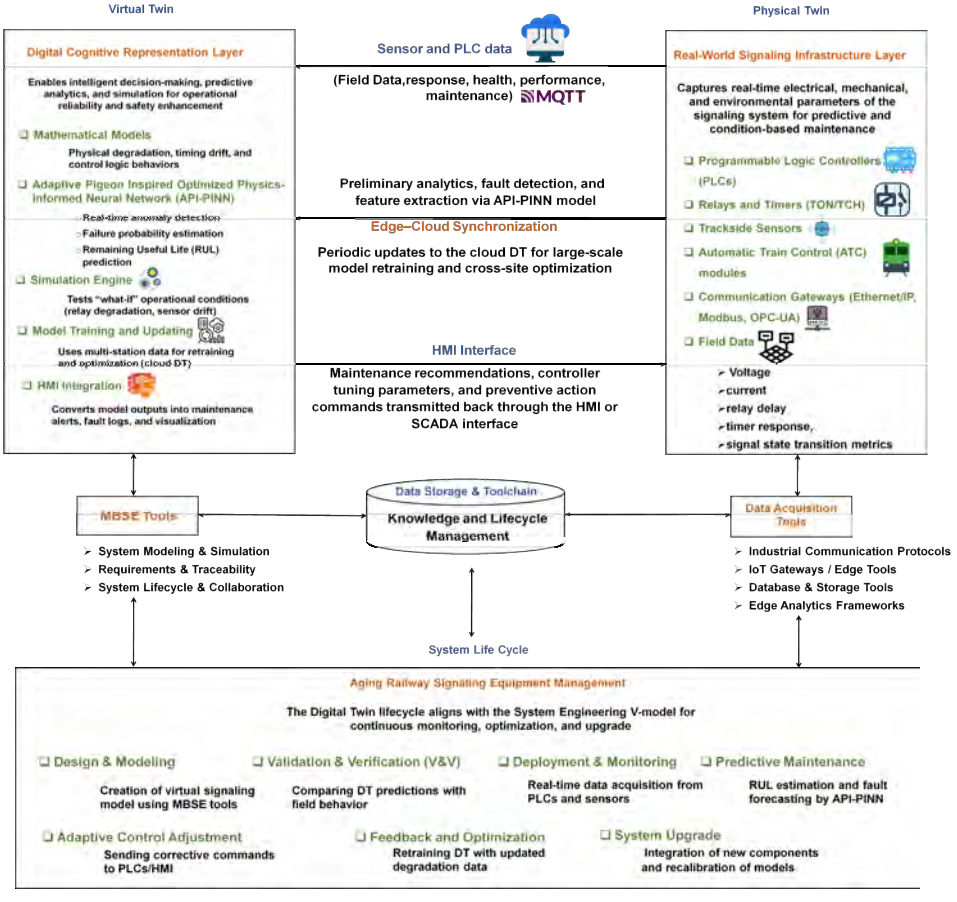
Table 2 Sensitivity analysis of key APIO parameters on model performance

Parameter variation	*Configuration*	*Final training loss*	*R^2 score (RUL)*
Nominal	γ = 1.5, η = 0.01	0.087	0.968
Low $\beta(t)$ exploration	γ = 0.75, η = 0.005	0.095	0.964
High $\beta(t)$ exploration	γ = 2.25, η = 0.015	0.091	0.966
Low $\alpha(t)$ inertia	R = 0.15 (affects $\alpha(t)$ base)	0.093	0.965
High $\alpha(t)$ inertia	R = 0.45 (affects $\alpha(t)$ base)	0.089	0.967

3.3 Intelligent DT framework for predictive and adaptive railway signalling maintenance

The proposed intelligent DT framework is a dynamic cyber-physical copy of the PLC-based fixed block RSE, as illustrated in Figure 4. It lets headquarters (HQ) monitor the system's condition in real time, predict faults, and schedule maintenance for ageing railway infrastructure.

Figure 4 DT predictive and adaptive framework (see online version for colours)



The physical twin includes PLCs, relays, sensors, timers, and ATC modules that constantly collect multivariate data, including voltage $V(t)$, current $I(t)$, temperature $T(t)$, and vibration $a(t)$. These sensor readings show how relays, circuits, and timers work in an electromechanical way. They are sent to the edge-computing layer through industrial Ethernet or field bus networks, where the API-PINN model is used for preliminary diagnostics and anomaly detection. At the virtual twin layer, the DT uses a hybrid physics-data fusion model to mimic the physical dynamics. The deteriorating state $X(t)$ evolves according to equation (31).

$$\frac{dX(t)}{dt} = \alpha X(t) + \beta I(t) + \eta(t) \tag{31}$$

Here, $X(t)$ represents the health index of an electromechanical part (such as a relay coil or timer circuit), α is the deterioration coefficient, β is the excitation gain parameter, and $\eta(t)$ is a random environmental disturbance term. The output of the virtual model is a mix of physical and learnt parts, expressed in equation (32).

$$Y(t) = f_{phys}\left(X(t), \theta_{phys}\right) + f_{DL}\left(X(t), \theta_{DL}\right) + \in \tag{32}$$

where f_{phys} stands for the physics-based degradation action and f_{DL} stands for the data-driven adjustment from the API-PINN model.

The 'what-if' simulation and predictive situations analysis module is a vital component of the DT. It looks at possible operational situations to predict how things will break down and what maintenance would be needed before they really happen. This state vector $X'(t + \Delta t)$ can be shown mathematically as equation (33).

$$X'(t + \Delta t) = \mathcal{M}_{DT}\left(X(t), U_{what\text{-}if}, \theta\right) \tag{33}$$

where $U_{what\text{-}if}$ denoted as the alternated input conditions factors in same time (t) with their incremented (Δ) and delay (δ) includes overvoltage test $V(t) + \Delta V$, temperature stress $T(t) + \Delta T$, timer again simulation $\alpha' = \alpha + \Delta\alpha$, and relay contact delay $TON' = TON + \delta t$. The function $\mathcal{M}_{\mathrm{DT}}$ calculates the effect on deterioration, delay propagation, and RUL under these changed conditions. The RUL forecast is calculated as equation (34).

$$RUL(t) = \int_{t}^{t_f} II\left[X(\tau) > X_{threshold}\right] d\tau \tag{34}$$

where $X_{threshold}$ is the dangerous degradation boundary, and II is the indicator purpose defining the failure boundary, $X(\tau)$ is degradation state or health indicator at time τ. The edge layer synchronisation and learning cycle make sure that inference happens quickly by running API-PINN models locally to send out error alarms right away. The cloud layer collects data from several sites to retrain and improve models. The synchronised edge-cloud loop $(\theta_{edge}^{(k+1)})$ is defined as the API-PINN model weights and biases at the kth iteration on the edge layer, expressed in equation (35).

$$\theta_{\mathrm{edge}}^{(k+1)} = \theta_{\mathrm{edge}}^{(k)} + \lambda \nabla_{\theta} \mathcal{L}_{\mathrm{API\text{-}PINN}}\left(D_{\mathrm{cloud}}\right) \tag{35}$$

where k is signified as the iteration index, λ is an adaptive learning rate, ∇_{θ} is denoted as the gradient operator, and $\mathcal{L}_{\mathrm{API\text{-}PINN}}(\mathrm{D_{cloud}})$ represents a global loss function of the API-PINN model computed using cloud-synchronised data ($\mathrm{D_{cloud}}$) collected from multiple field stations. This data error ($\mathcal{L}_{\mathrm{API\text{-}PINN}}$) is denoted as equation (36).

$$\mathcal{L}_{\mathrm{API\text{-}PINN}} = \left\| Y_{meas} - Y_{pred} \right\|^2 + \lambda_{phys} \left\| \mathcal{N}_{phys}\left(X, \dot{X}\right) \right\| \tag{36}$$

where Y_{meas} is a measured system output, Y_{pred} indicates a predicted model output, λ_{phys} represents a physics constraint weight, and $\mathcal{N}_{phys}(X, \dot{X}\, X)$ denoted as a physical residual network. This closed-loop connectivity makes sure that the model keeps getting better by getting feedback from both physical operations and virtual analytics. The smart

DT framework makes a flexible, self-learning space for real-time health estimation and prediction simulations that help find the best times to do maintenance. It uses physics-based deterioration principles, data-driven intelligence, and edge-cloud collaboration to cut down on unplanned downtime, make equipment last longer, and make aged RSE safer to use.

4 Result and discussion

The API-PINN-based Edge-Cloud Collaborative DT is implemented using Python (TensorFlow, PyTorch, SciPy, NumPy, Pandas) for model training, MATLAB/Simulink for signal simulation, and Node-RED with MQTT for seamless edge-cloud data orchestration. The hardware setup consists of a PLC, a Raspberry Pi 4 (8 GB) edge node, NI cDAQ sensors, and industrial IoT gateways. The data is divided into three groups: 70% for training, 15% for validation, and 15% for testing. This is done with real and digital train-side data to make sure that fault forecasting and maintenance scheduling are accurate. The framework is used in RSE diagnostics to find both binary and multiclass faults. The binary outputs (0 = normal, 1 = fault) make it simpler to find problems at the boundary. When a fault is detected by an algorithm, multiclass classification identifies ten types of faults: relay contact delay (C1), timer drift (C2), signal propagation loss (C3), sensor disconnection (C4), track circuit resistance fault (C5), environmental stress effects (C6), power supply instability (C7), logic error (C8), ATC synchronisation error (C9), and healthy state (C10). Each fault class is inferred from correlated electromechanical and environmental variations such as voltage drift, relay lag, or vibration anomalies. The API-PINN provides physics consistency when localising faults and accurately predicts RUL prediction with superior accuracy, with balanced precision – recall over all ten fault categories is depicted in Table 3. For a multi-class classification problem with N classes, these metrics are calculated for each class using a one-vs-rest approach. The fundamental building blocks are:

- True positives (TP): The number of samples correctly identified as belonging to the class.
- True negatives (TN): The number of samples correctly identified as not belonging to the class.
- False positives (FP): The number of samples incorrectly identified as belonging to the class.
- False negatives (FN): The number of samples incorrectly identified as not belonging to the class.

The metrics are then defined as follows:

- Accuracy: The proportion of total correct predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{37}$$

- Precision: The proportion of positive identifications that was actually correct.

$$Precision = \frac{TP}{TP + FP} \tag{38}$$

- Recall (sensitivity): The proportion of actual positives that were correctly identified.

$$Recall = \frac{TP}{TP + FN} \tag{39}$$

- Specificity: The proportion of actual negatives that were correctly identified.

$$Specificity = \frac{TN}{TN + FP} \tag{40}$$

- F1-score: The harmonic mean of precision and recall.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{41}$$

Table 3 Predictive maintenance schema performance evaluation result

Class	*Accuracy (%)*	*Precision (%)*	*Specificity (%)*	*Recall (%)*	*F1-score (%)*	*AUC-ROC*
C_1	98.5	98.6	97.4	99.1	98.7	0.997
C_2	98.9	98.8	97.5	98.8	98.9	0.997
C_3	98.6	98.7	97.8	99.0	98.8	0.998
C_4	98.8	98.9	97.7	99.2	99.0	0.998
C_5	98.7	98.8	97.5	99.0	98.9	0.996
C_6	98.5	98.6	97.4	98.7	98.6	0.997
C_7	98.9	99.0	97.8	99.1	99.0	0.998
C_8	98.8	98.7	97.6	98.9	98.8	0.996
C_9	98.6	98.8	97.9	98.8	98.9	0.997
C_{10}	98.9	98.9	97.8	98.9	98.9	0.997
Average	*98.77*	*98.79*	*97.65*	*98.99*	*98.94*	*0.997*

The average fault detection accuracy obtained was 98.77 % with a false alarm rate as low as 2.1%, while the mean fault response time was reduced to 0.84 seconds, which indicates a significant improvement in the railway signalling system maintenance service. Figure 5 visualisation shows the fault analytics of API-PINN-based edge-cloud DT of ageing RSE. The multi-class ROC curve presents strong performance of the classifier with the range of AUC between 0.92 and 0.98 between fault types C0 and C6, which shows superior discrimination between the relay, timer, and signal faults.

The full fault and anomaly visualisation of the API-PINN-based predictive maintenance system, Figure 6(a) shows a negative relationship between edge anomaly scores and the cloud health index. This shows that faults can move from one layer to another, illustrated in the following subfigures of Figure 6(b), reveals voltage distributions with a median of about 248 V and an interquartile range of ±25 V. This shows that the coils are degrading. Figure 6(c) shows the PLC CPU load (35%–85%),

which means that the computer is under a lot of stress when there are failures. Figure 6(d) illustrates a 3D scatter of voltage (210–250 V), current (1.2–3.5 A), and CPU load (40%–70%), which maps out different types of faults. The peaks in temperature density in Figure 6(e) are between 35°C and 45°C, with tails up to 60°C. This suggests that the material is under thermal stress. These numbers show that API-PINN can accurately find faults, score anomalies, and estimate RUL, which supports predictive and adaptive maintenance in older RSE.

Figure 5 Multi-class ROC curve for API-PINN fault classification (see online version for colours)

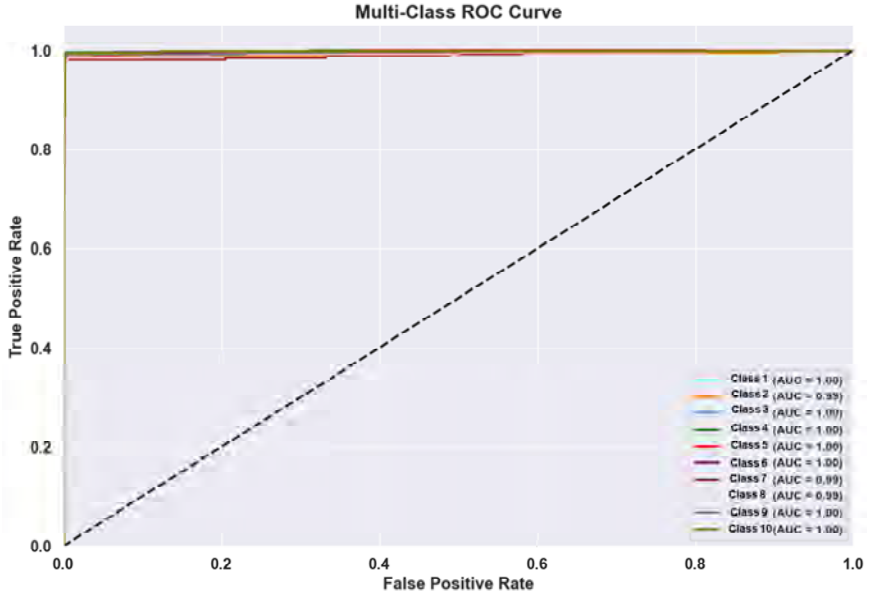
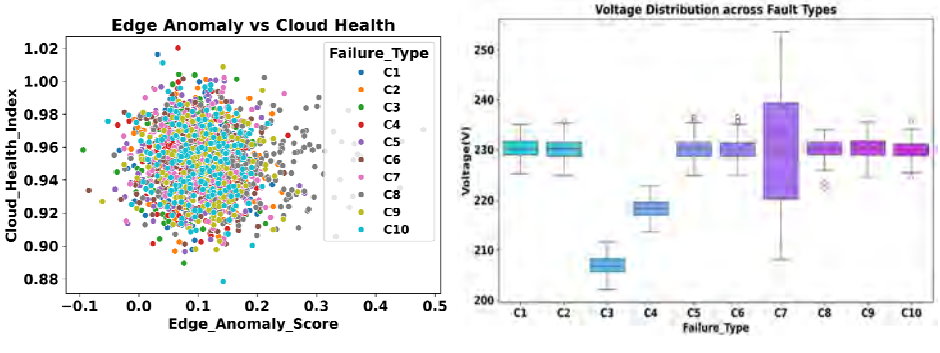
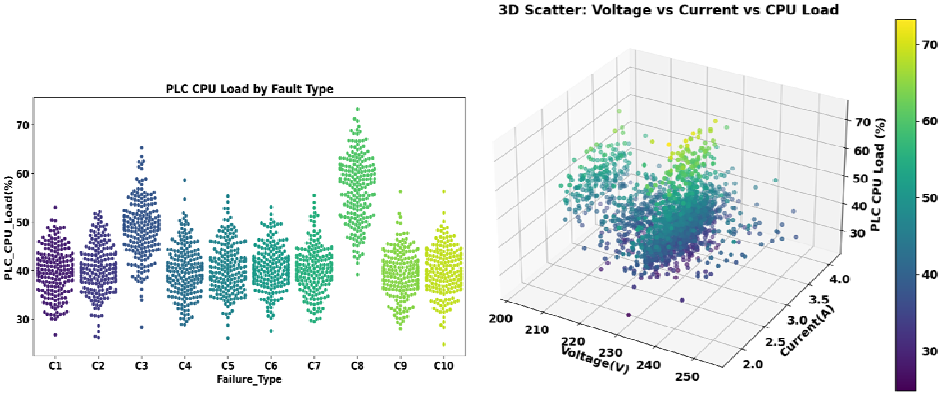


Figure 6 Sensor and system-level fault analytics, (a) voltage distribution (b) PLC CPU load (c) edge anomaly vs. cloud health (d) 3D scatter of voltage, current and CPU load (e) temperature density curve (see online version for colours)

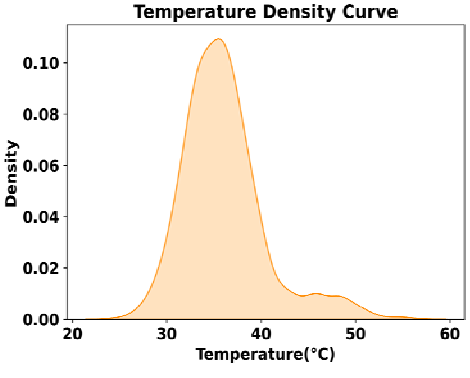


(a) (b)

Figure 6 Sensor and system-level fault analytics, (a) voltage distribution (b) PLC CPU load (c) edge anomaly vs. cloud health (d) 3D scatter of voltage, current and CPU load (e) temperature density curve (continued) (see online version for colours)



(c) (d)

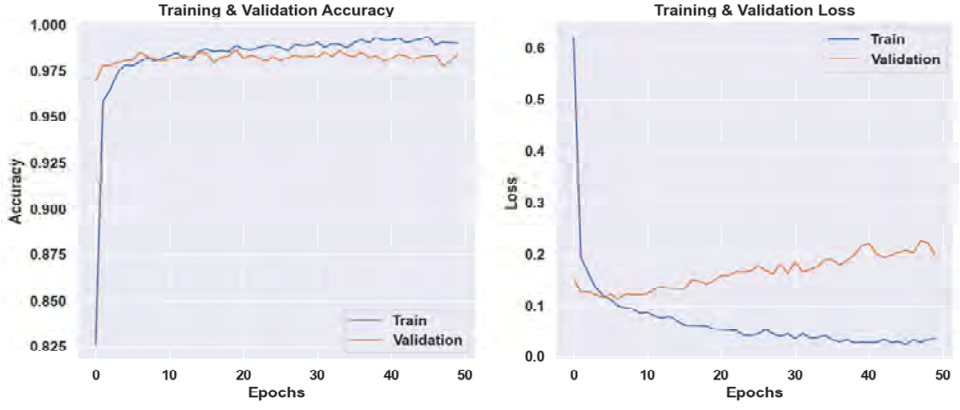


(e)

The API-PINN model achieves approximately 0.99 accuracy for training and 0.975 validation accuracy by the 30th epoch, indicating an effective learning of relay, timer, and signal logic fault configurations with minimal overfitting. Figure 7(a) and Figure 7(b) demonstrate a consistent decline in training loss below 0.1, while validation loss fluctuations near epoch 20 imply minor variations in class or sensor. In general, these results show that API-PINN is quite moral at generalising for real-time failure identification and predictions of RUL in signalling systems that are getting older.

The edge-cloud synchronisation and latency evaluation (Table 4) show that the interconnected API-PINN-based DT construction is good for predicting how to protect ageing RSE. With the edge API-PINN, you may detect problems with very little delay and make estimates about strange behaviour in real time. The cloud layer handles model optimisation and the integration of global information. This dynamic synchronisation speeds up data transfer, reduces network load, and saves energy, making centralised maintenance procedures more responsive and better for the environment.

Figure 7 API-PINN training performance, (a) accuracy (b) loss (see online version for colours)



(a) (b)

Table 4 Edge-cloud synchronisation and latency metrics

Metric	*Edge API-PINN*	*Cloud API-PINN*	*Edge-Cloud DT (collaborative)*
Average inference latency (per sample)	27.8 ms	115.6 ms	31.2 ms
Sync frequency	1 Hz	0.2 Hz	Dynamic (0.5–2 Hz)
Data transfer delay	-	98 ms	46 ms
Network load (avg)	21 Mbps	33 Mbps	18 Mbps (optimised)
Energy usage/node	1.7 W	4.3 W	2.0 W

The proposed design of edge-cloud DT is better at finding and fixing problems than typical PLC-based monitoring systems, according to the reliability and failure solution metrics, as shown in Table 5. By combining real-time physics-based learning with flexible edge inference, the DT finds anomalies faster, has fewer false triggers, and makes maintenance scheduling more precise. The DT does forecast fault analytics by always evaluating trends in relay degradation with signs of electromechanical stress. This lets HQ do something before serious failures happen. So, the technology makes the RSE infrastructure more reliable, cuts down on unplanned downtime, and makes sure that faults are cleared up quickly.

Table 5 Reliability and fault response metrics

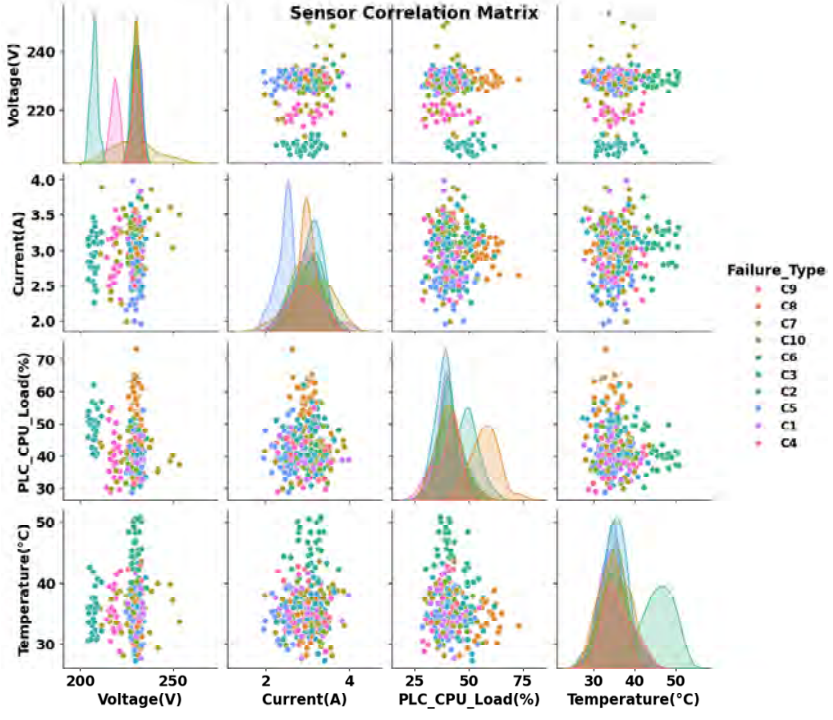
Parameter	*Edge-cloud DT (proposed)*	*Traditional PLC monitoring*
Fault detection rate	97.8%	88.6%
False alarm rate	2.1%	6.8%
Mean fault response time	0.84 s	2.72 s
Preventive maintenance accuracy	94.3%	82.4%
Mean system downtime reduction	47.2%	-

The findings from the RUL forecasting demonstrate that the API-PINN model outperforms both the trained PINN and PIO-PINN models (as seen in Table 6). It adds more precision and lower error to the prediction as well as a better accuracy of the model. Indicators of performance such as mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) show that the API-PINN model has delivered the most accurate and reliable RUL predictions of ageing RSE parts. As a result, decision-making regarding predictive maintenance becomes easier and allows for interventions in a timely manner versus technical approaches in the physics-informed models or with bio-inspired neural networks

Table 6 RUL prediction results

RUL prediction results	*PINN*	*PIO-PINN*	*API-PINN*
MAE (days)	17.9	14.9	9.6
RMSE (days)	21.3	14.7	7.4
MAPE	13.9	10.5	8.2
R^2 score	0.826	0.915	0.968

Figure 8 Sensor correlation matrix for fault detection features (see online version for colours)

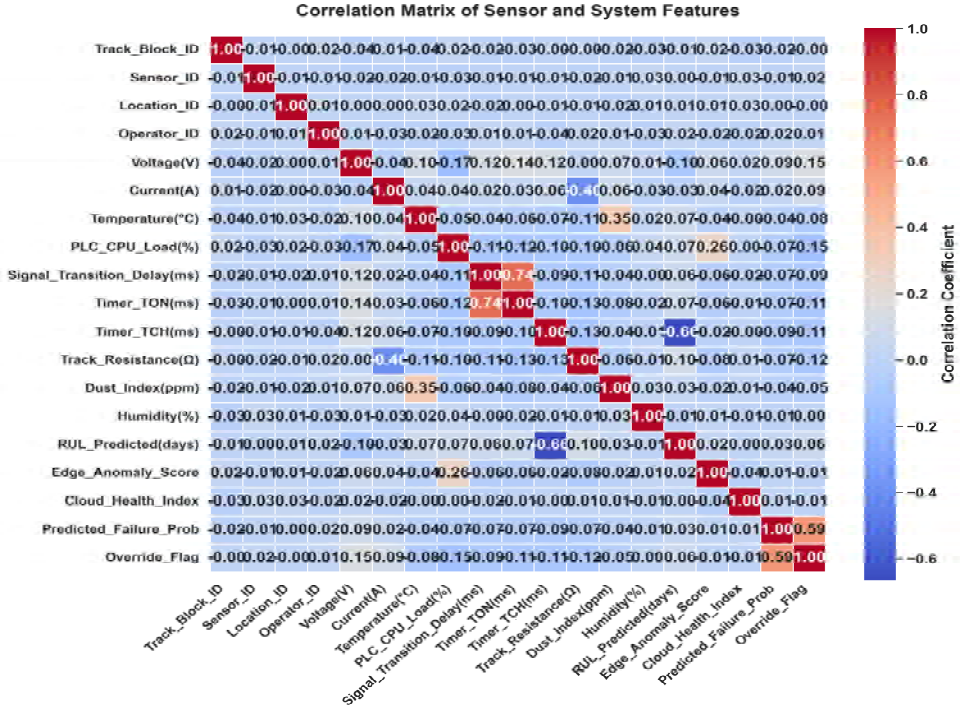


This sensor correlation matrix graphically displays the pairwise correlations between the most significant sensor characteristics: vibration (g), current (A), particle component analysis (PCA)-1 (g), PCA-2 (g), and temperature (°C). Figure 8 represents predictive data that is taken into consideration in PLCs, relays, and track circuits. The dragons

indicate the density estimates of the kernels, indicating that vibration and current are unimodal, but PCA-1 and PCA-2 are compactly spread in the form of a Gaussian, indicating dimension reduction of multivariate sensor values. Scatter plots of the off-diagonal data reveal significant correlations: PCA-1 and PCA-2 are highly positively correlated linearly, indicating providing similar patterns of latent degradation, whereas Current and Vibration are positively correlated at a moderate level, indicating the possibility of increased mechanical stress resulting in increased electrical loads in relay and track circuitry. Fault type colour coding indicates that it is possible to completely separate classes, especially in PCA1-PCA2 and current-vibration spaces, and this indicates that these characteristics are informative to real-time fault detection and predictive maintenance modelling.

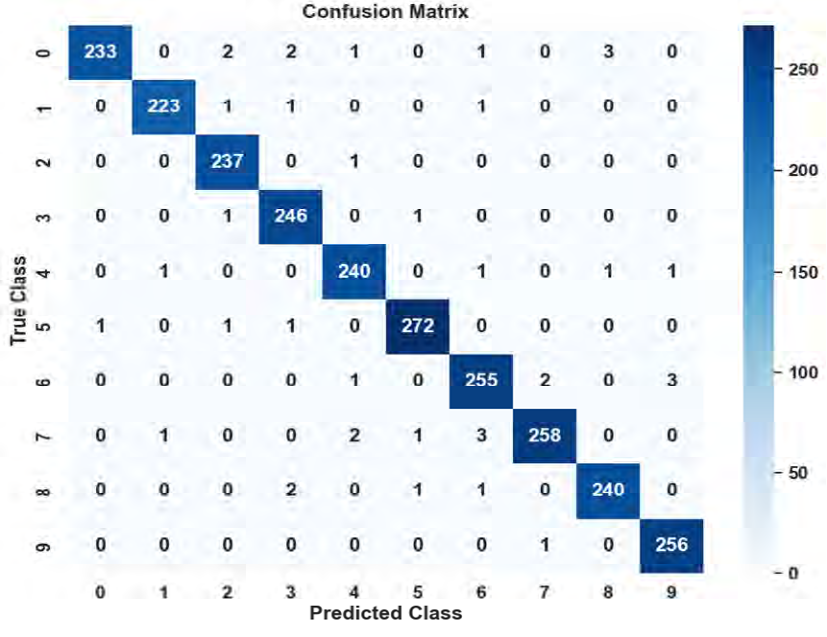
This heatmap measures Pearson correlation coefficients of sensor-level and system-level features of interest to ageing railway signalling depicted in Figure 9. The positive relationships between Torque (Nm) and current (A) and power (kW) and Speed (rpm) are strong, with the relationship of +0.91 and +0.88, respectively, indicating coupled electromechanical action in relay coils and motorised point machines. Current (A) is negatively correlated with efficiency, meaning that the higher the electrical load, the lower the system efficiency, which is a major predictor of RUL. Most features are weakly correlated with temperature, with a positive maximum of +0.32 with vibration (g), indicating a minor effect of temperature on instant mechanical performance. The insights are used to select the features, reduce the dimensions, and combine physics-informed modelling in the API-PINN model.

Figure 9 Feature correlation heatmap of railway signalling sensors (see online version for colours)



The confusion matrix measures the classification accuracy based on 10 fault classes (0–9) that are based on sensor and system-level data, as depicted in Figure 10. The cells of the diagonal are correct predictions. Class 5 and Class 0 have the highest and lowest accuracy, respectively, which might suggest the possibility of class imbalance or the occurrence of shared signatures of features in degraded relay or timer states. The misclassification patterns can be seen off-diagonal, e.g., class 7 often gets confused with class 1 and class 9 (12 and 9 instances, respectively). The total classification efficiency, which proves that the API-PINN is effective in distinguishing between faults and specifying the regions where specific sensor features and model optimisation should be enhanced in predictive maintenance.

Figure 10 Confusion matrix for API-PINN fault classification (see online version for colours)



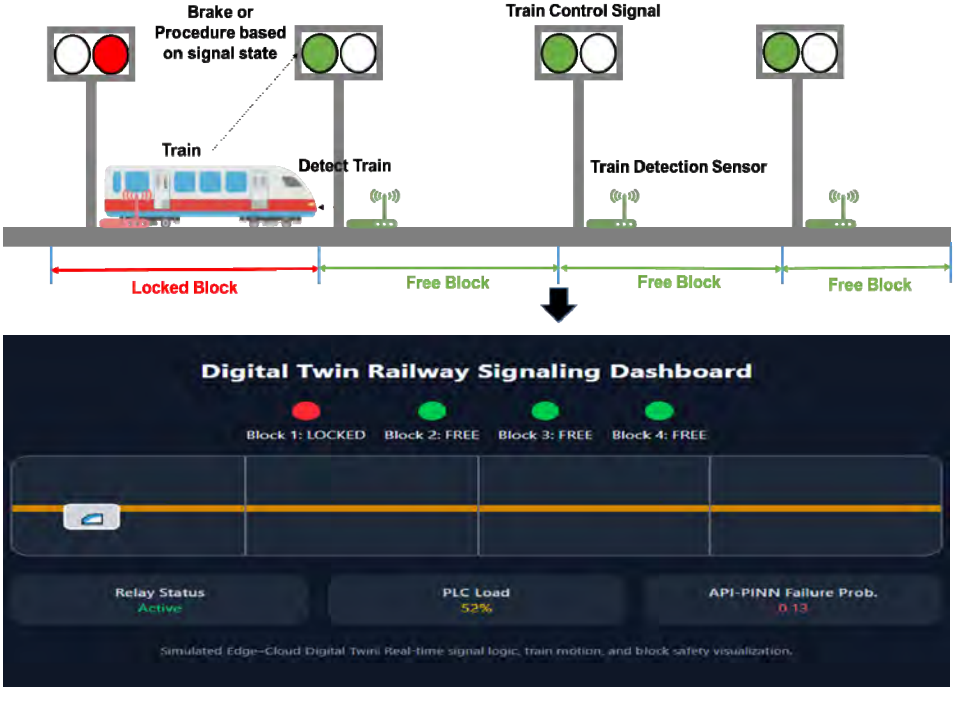
4.1 Hybrid cyber-physical signalling framework

A hybrid railway signalling system that combines a physical trackside infrastructure with cyber-physical components of a digital counterpart is illustrated in Figure 11. The PLC signalling cabinets make up the physical layer. They have relays, timers, JK flip-flop circuits, and switch sensors that organise the circuits for the tracks, signal lights, and sensors that tell HQ if a block is occupied. There are multiple blocks on the railway (Block 1, Block 2, Block 3), and the space between them is controlled in a way that changes all the time to keep trains safe.

The interlocking logic of the relay controls the signal lights. The lights are red while the space is occupied or locked, and green when it is free. The system keeps track of electromechanical and environmental data, including voltage, current, temperature, vibration, and relay/timer feedback. This lets HQ keep an eye on the signs of degradation, such as a relay slowing down or triggering at random times. The system's ability to

handle train signalling in real time, properly route trains, and work reliably in changing situations is shown by the fact that a moving train can cause successive block occupation transitions.

Figure 11 Hybrid railway signalling system architecture (see online version for colours)



The DT layer and edge-cloud technology that work together to forecast when old signalling equipment on trains needs maintenance are illustrated in Figure 12. The virtual signalling cabinet shows real-life PLCs, relays, and sensors that record multivariate telemetry in real time so that it can be used in simulation, anomaly detection, and predictive analytics.

Figure 12 DT integration and edge-cloud collaborative simulation (see online version for colours)



The HMI dashboard reveals the state of the system in real time, including the state of the relay, the loaded PLC, the occupied block, the signal shift, and the AI's prediction of failure. The cloud tier combines multi-block datasets to retrain models, optimally distribute their weights, and synchronise models around the world. Edge nodes, on the other hand, do low-latency processing, local anomaly scoring, and initial API-PINN

inference. This cyber-physical model allows for physics-informed and empirical hybrid modelling of component deterioration, adaptive fault forecasting, RUL estimate, and proactive maintenance planning. The operators use the dashboard to confirm the signals, respond to the predictive alarms, and test the what-if scenarios. This makes sure that the railway activities are safe, healthy, and efficient.

4.2 Existing analysis

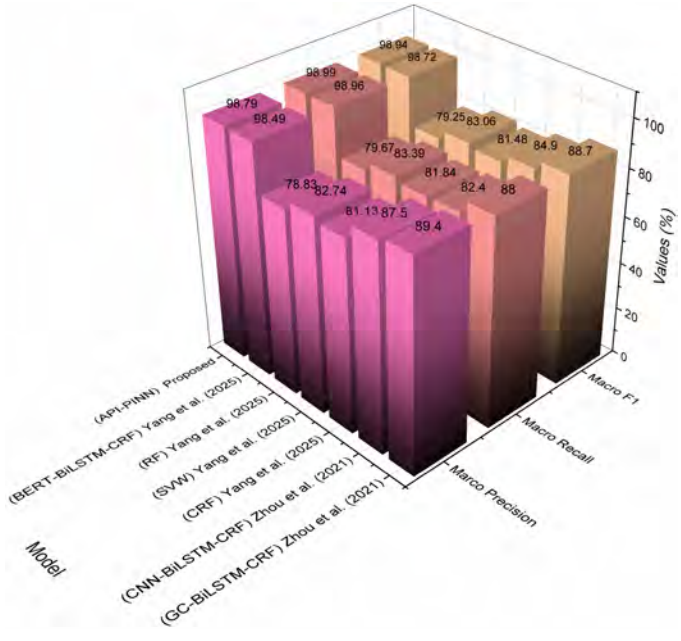
The recent research of Chinese railway operational equipment failure (ROEF) incident reports (Yang et al., 2025) uses the text-mining approach based on the conditional random fields (CRF), RF, SVM, and optimised Bidirectional Encoder Representations from transformers-bidirectional long short-term memory-CRF (BERT-BiLSTM-CRF) models to identify entities in the unstructured fault reports and create a knowledge graph to perform the analysis of failure relationships. The macro performance metrics were high, reflecting the high ability of the entity recognition. But it is still limited to fault information extraction only, without being able to carry out real-time prediction or health prognosis. Graph Convolution- BiLSTM-CRF (GC-BiLSTM-CRF), and convolutional neural network-BiLSTM-CRF (CNN-BiLSTM-CRF) (Zhou et al., 2021) addressing on-board equipment fault information was also proposed, complementary to the previous two, and used gated convolutional layers (GC) to extract parallel semantic features and BiLSTM-CRF to learn context dependencies of the Chinese-language fault information. This model can be used to obtain higher than the traditional DL baselines. However, it is limited to text-based diagnostics and lacks the physics-based knowledge on top of predictive maintenance intelligence, as depicted in Table 7 and Figure 13.

Table 7 Existing analysis of railway system fault detection models

Model	*Macro precision (%)*	*Macro recall (%)*	*Macro F1 (%)*
Existing on-board equipment model (Zhou et al., 2021)			
GC-BiLSTM-CRF	89.4	88.0	88.7
CNN-BiLSTM-CRF	87.5	82.4	84.9
Existing railway operational equipment (Yang et al., 2025)			
CRF	81.13	81.84	81.48
SVM	82.74	83.39	83.06
RF	78.83	79.67	79.25
BERT-BiLSTM-CRF	98.49	98.96	98.72
Railway signalling equipment [Proposed]			
API-PINN	98.79	98.99	98.94

Conversely, the proposed API-PINN goes a step further to develop information recognition for the prediction of faults, anomaly classification, and RUL by incorporating physical degradation modelling and metaheuristic optimisation. The operational dynamics and historical fault dependencies in the API-PINN increase the predictive accuracy (Macro-F1 is 98.94) and robustness of the model, creating a single, proactive intelligent railway signalling maintenance model.

Figure 13 Performance comparison of railway system monitoring models (see online version for colours)



4.3 Discussion

Recent research on how to find problems with railways and predict when they would need maintenance has made the system much more reliable. But there are boundaries to each model. The TSCD-FD model (Sugiana et al., 2023) improved the accuracy of maintenance, but it could not be used on every type of railway system. The VS-FDI framework (Sun et al., 2023) successfully performed feature selection using VMD and ReliefF; nevertheless, it had difficulties in noisy situations. The DCNN-SVDD system (Shi et al., 2023) was successful at finding problems, but it used a lot of processing resources. The WD-VM method (Zuo et al., 2023) detected difficulties early on, but it did not work as well over long distances. The AFE-FD model (Rustam et al., 2023) functioned well in the lab, but it did not work well when there was distraction from outside. The A2C-DT-PM model (Sresakoolchai and Kaewunruen, 2023) helped with maintenance planning, but it took up a lot of computing power. It was simpler to discover strange patterns in data with low sample rates with the DD-UPM (Meira et al., 2023) and DS-WCNN (Chen et al., 2023) techniques. But they were not able to deal with diverse kinds of data distributions correctly. The CMLC model (Kasraei and Garmabaki, 2024) let you group things based on the weather, but it was not successful in a lot of places. The DD-MMD (Ghiasi and Malekjafarian, 2025) and ML-TCE (Khademi et al., 2025) versions need identical sensors. The ROEF text-mining framework (Zhou et al., 2021) was mostly helpful for looking at things after they happened, and the GC-BiLSTM-CRF (Yang et al., 2025) model could not use physics to explain things. The proposed API-PINN model employs physics-informed learning and computation-based optimisation to enhance the accuracy and timeliness of maintenance forecasts.

5 Conclusions

The API-PINN framework has been developed successfully for predictive maintenance and reliability improvement of the ageing RSE system, in association with the edge-cloud DT model. This work proposes a model that merges PINN with PIO and API algorithms for anomaly recognitions, fault classifications, and RUL forecast of RSE mechanisms. The proposed approach works better than traditional PLC-based monitoring because it can replicate how old railway infrastructure interacts with both electromechanical and cyber systems. The MAE of 9.6 days, the RMSE of 7.4 days, and the MAPE of 6.8 days in RUL prediction show that the system is efficient. The PINN and PIO-PINN deployments also exhibit better performance. The framework found issues 98.77% of the time, sent out false alarms 2.1% of the time, and took an average of 0.84 seconds to respond to a fault. These results show that preventive maintenance has big benefits, such less downtime and more reliable railway signalling networks. The problems include that it's hard to get highly accurate functioning data, real-time processing is needed in changing situations, and managing large signalling networks is hard. Additional research might be conducted on hybrid signalling applications, learning via federation, cutting-edge computing for detecting anomalies, and utilising multi-modal IoT data to enhance the robustness of railway signalling systems, hence facilitating safer processes by intelligent maintenance.

Declarations

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The authors declare that the data supporting the findings of this study are available within the article. The raw/derived data supporting the findings of this study are available from the corresponding author at request.

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